# Ibasho agent blackboard

## State (orchestrator, kept current)

**orchestrator** state · active
**Phase:** M-A first light plus independent M-B work, light testing. Full history: `docs/archive/BLACKBOARD-to-M0.typ`.
**Done:** G0 scaffold, M-0 slice (physics, rules, search and money, Lean spec and checker with TS mirror, narrator and memory, scene, agent link, first app screen, content and landing, hero assets), SPEC 14 contracts in core.
**In progress:** WP-01 physics (A1 to A10, maintenance decay), WP-06 UI (M-A and M-B components), WP-07 scene (review), WP-08 app (shell, onboarding, Reveal with scene), WP-03 rules (M-B), WP-02 money and score (M-B, SPEC 14 run and negatives), WP-04 checker gaps (bills, score, memory.completed).
**Idle until M-D:** WP-05 AI, WP-09 content, WP-12 agent link, WP-10 QA (testing is light for now), WP-11 assets.

## Contract index (where the source of truth lives)

- Contracts and zod schemas: `@ibasho/core` (`packages/core/src/**`, `@ibasho/core/schemas`). SPEC 14 kinds: `maintenance`, "If nothing is done" run, per-package `negatives`.
- Measure and option ids: `packs/home-uk/src/ids.ts` (frozen; GH1 432 heavy, 10,800 packages).
- Entry points: packs export `./engine`, `./ui`, `./ids`; apps start at `src/main.tsx`; `packages/fixtures` exports `./*` (no index).
- Certificate, constants, seals, spec hash: generated from Lean into `@ibasho/spec-gen` (real, not a placeholder).
- Ownership as code: `tools/lint/ownership.ts`. `ACKNOWLEDGEMENTS.md`, the blackboard, STATUS and `docs/refs.yml` are shared append.
- Agent link: `node apps/mcp/src/main.ts` (no build); `pnpm agent:install`.

## Decisions (condensed; details in the archive)

**orchestrator** decision · final

- Stack: pnpm monorepo, Vite, React, TypeScript 6.0.3, R3F, IndexedDB, Biome, Lean 4 v4.34.0 core only. Local first, offline first. Versions in ARCH 13.
- Plan of record: VISION, SPEC (rulings R1 to R24), ARCHITECTURE, DESIGN_SYSTEM (Andon), BUILD_PLAN; red-team rulings in `docs/review/DECISIONS.md` override older text.
- Hero home GH1: 1905 London mid-terrace in a conservation area. Score weights F 1/5, C 1/5, E 3/20, A 3/20, H 1/10, R 1/10, T 1/10, from Standard mode; shown as a number with four groups, no letter. Futures: Least upfront, Best return, Lowest running carbon.
- Lean owns formulas and predicates; pack data travels in certificates; the checker is proved sound and the TS mirror matches it.
- Engine determinism: no inexact `Math` in engine code, `dmath` instead, fixed point after quantise.
- Narrator is a plugin: template always on, Ollama local (`qwen3:8b` default; `:cloud` models count as cloud and go in the outbound ledger), Claude later.
- Agent link (WP-12): MCP server, bridge mounted by the companion, Claude Code plugin, installer. Companion: loopback host, Origin allow list, Bearer token for scripts.
- Bad outcomes (SPEC 14): "If nothing is done" line, custom what-if requests as typed ops, negatives beside savings.
- Reuse proven open source and credit it in `ACKNOWLEDGEMENTS.md`. Asset licensing is not a constraint.
- Testing is light for now: each WP runs only its own package tests. No gates until the owner asks.

**orchestrator** decision · final
Owner direction (2026-10-11): prototype first, outside in. A high-fidelity, responsive prototype of every user flow (new, revisit, restart, current plus what-if, update) with an honest staged progress bar for rendering and computing, on fixture data shaped like real engine output and labelled Preview. Then replace fixtures with real computation stage by stage. See `docs/research/v3/00-owner-direction.md`.

**director** decision · final
v3 plan of record (2026-10-11): `docs/PLAN-V3.md`, with SPEC 15 to 26 (rulings R25 to R45), ARCH 15, BUILD_PLAN 9 to 12 and DESIGN_SYSTEM 13. A studio for eight building types in a real 3D city, Hyderabad

first (hero: Vindhya Bhavan, IIIT). Order: V-0 contracts (WP-00, WP-10), then wave 1 V-P prototype on Preview data (design WP-06; developers WP-08, WP-15, WP-07, WP-13, WP-14, WP-17; QA WP-22 product, WP-23 UX, WP-24 content, review based), then real stages V-1 to V-6 in progress-bar order, then V-7 partner and V-H hero. New WPs WP-13 to WP-24 own disjoint paths (BUILD_PLAN 10). Testing stays light.

**orchestrator** decision · final
Room names (owner request for find by name, 2026-10-11): fixtures may carry institutional or functional room names from the drawings; never people's names. One label rule in plan and 3D: function first, then name. PLAN-V3 section 8 updated; studio review A-M1 resolved by this ruling.

**orchestrator** decision · final
Owner request (2026-10-11): proactive sustainability. New sixth flow "Design before you build" (SPEC 27): brief with typed goals, design rooms and floors, place on a real map (London, Hyderabad), compare plots and orientations, shape to goals with the optimiser, commit as a branch. Owners: B flow and editor, C placement and site comparison in the scene, A richer London area, UK new-build code packs (Part L 2021, Part O, Future Homes Standard, London Plan) and goal targets. Queued behind the current studio review fixes.

## Claims

**steward:WP-00** claim · done
SPEC 14 contracts in `packages/core/**`: the `maintenance` plugin kind, the "If nothing is done" run, per-package negatives, and the deferred WP-02 `OwnerCtx` fields.

**builder:WP-08** claim · active
Gate M-A (app shell and store): `apps/web/**` except `apps/web/public/assets/**`, `packages/store/**`. Shell layout zones (left rail, top bar with home switcher and lantern, mobile tab bar), `/app/start` onboarding in 5 typed steps with no budget question, Reveal on the home stage with WP-07 `Stage` (SVG fallback without WebGL), packs imported by package name.

**builder:WP-06** claim · active
Gates M-A and M-B UI kit: `packages/ui/**`. M-A: `Sheet`, `NarratorStrip`, `ScoreGroups`, `EpcChip`. M-B: `Futures` (with the fourth line "If nothing is done" and a negatives list on every card), `MeasureCard`, `FuelBars`, `Slider`, `WhyNot`, `CheckLane`, `MonthRing`, plus the chart frame with `TableAlt`, a first mon set and `GradeBadge`. Stories for each.

**builder:WP-02** claim · active
M-B pieces without new physics: score with exact Q and the four groups, EPC band proxy, IRR, paybacks, loan stub, flips, the "If nothing is done" run and per-package negatives (SPEC 14), costs for every SPEC 9 option. Paths: `packages/engine/src/{api,index}.ts`, `packages/engine/src/{search,finance,score,flips,evidence}/**`, `packages/engine/{test,scripts}/**`, `packs/money-uk/**`, `packages/fixtures/src/results/**`.

**builder:WP-03** claim · done
M-B (C1 to C7): `packages/engine/src/rules/**`, `packs/rules-uk/**`, `packs/rules-in/**`. Named functions, `noise()`, the SPEC 9 `rules-uk` set with `APPL.*`, sentences, obligations, rule lint.

**agent-A:WP-00** claim · active
**v3 platform and data (Agent A).** WP-00 V-0 contracts (`packages/core/src/v3/**`, schemas, layer rules, installs), WP-10 `tools/lint/ownership.ts` v3 map, WP-13 `packages/preview` and `packages/fixtures/src/preview`, WP-14 `packages/plan` and `packages/fixtures/src/buildings`, WP-17 `packs/city-hyderabad`, `packs/city-london`, `tools/city`. Not touching B (ui, web, store, editor) or C (scene, packs ui halves).

**builder:WP-06+WP-08+WP-15 (agent B)** claim · done
**V-P prototype, product UI.** `packages/ui/**` (v3 kit: StageBar, PreviewStamp, ScaleLadder, LensPanel and AnalysisLegend, eight type cards and mons, EuiGauge, EndUseBar, RatingLadder, LedgerRow, CompareGrid,

FilmStrip, ModeTabs), `apps/web/**` except `public/assets` (studio shell, modes Make Look Choose Prove Tell, the five flows on Preview data), `packages/store/**` (Dexie v2 with migration), `packages/editor/**` (new, WP-15: plan editor). Builds against `@ibasho/core` v3 contracts in the working tree; Preview results come from a local adapter in `apps/web/src/studio/preview/` until `packages/preview` lands, then switch is one import.

**builder:WP-07 (agent C)** claim · done
V-P city scene: `packages/scene/src/city/**` (new `CityStage`, stand in campus context and Vindhya stand in, sun and time, lenses, penumbra, phototropic variant, lanterns, SVG fallback), `packages/scene/dev/**`, `packages/scene/test/**`. Reads `@ibasho/core` v3 contracts only; does not edit core, fixtures or city packs.

**agent-D:WP-09** claim · active
**v3 words and landing (Agent D).** `apps/landing/**` and `packages/content/**` only. Landing rewritten for the v3 studio (chapters gated by stage providers as well as plugins), v3 glossary and studio microcopy, voice guide update with a customer copy checker, building type copy.

**agent-E:WP-16** claim · active
**Site and daylight, first real analysis stages (Agent E).** `packages/site/**` and `packages/fixtures/src/site/**` only. SPA port, EPW weather (Hyderabad TMYx, London PVGIS TMY), Perez sky on Tregenza and Reinhart patches, CPU visibility kernel, sun hours, SVF, radiation, shading ledger, ASE and sDA (simplified, T1), real `sun`, `radiation` and `daylight` providers with the Preview contract shapes.

## Interfaces (live only)

**builder:WP-02** interface · open
**Requests (non-blocking):** (1) WP-03: `BUILTIN.MEMORY` lives in the search (`packages/engine/src/search/run.ts`) because the evaluator has no memory facts; move it into `BUILTIN_RULES` when a fact form exists, keeping the id. The budget rule receives the gross central cost as `p.capex`; say if Lean `Budget.lean` should compare N0 instead. (2) WP-01: heat pump costs size the unit with a cost-side estimate (`heatPumpSizeKw` in money-uk); expose `hp.size_kw` to cost plugins at M-A, e.g. a `size` argument or a twin field. (3) WP-00: `OwnerCtx` needs fuel escalation scenarios and a confirmed counterfactual replacement (A_cf) by M-B; M-0 uses 0% escalation and no A_cf.

**builder:WP-09** interface · open
**Chapter `requires` convention:** a requirement is a plugin id or `kind:<PluginKind>` (any plugin of that kind, checked with `registry.all(kind)`); `presenceOf(registry)` adapts a `PluginRegistry`. Current landing requires: drawn `climate-london` + `kind:archetype`; what-if `loft`, `ashp`, `kind:tariff`; futures `kind:search-strategy`, `rules-uk`; proof `kind:checker-rung`. **Request to WP-08:** the landing discovers plugins at build time from `packs/*/src/{engine,ui}/module.ts` (`apps/landing/src/discover.ts`), because `apps/web/src/manifest.ts` is still empty. Once the manifest lists the registered modules, please keep it importable from Node (plain `.ts`, no JSX) or export the module list as data, and WP-09 will switch to it so the page never advertises a pack the app does not load. **Links:** "Step into your home" goes to `/app/start`; "See a sample home" goes to `/app/home/gh1` (GH1 read-only, Story mode). WP-08: tell me if the sample route differs.

**builder:WP-09** interface · open
**To WP-10 (smoke):** `apps/landing` now prerenders the real page: `tests/smoke/boot.spec.ts` will fail on `main[data-andon="blank"]` being empty. Please assert `main[data-andon="landing"]` visible, `section[data-chapter="hero"]` present, title `Ibasho`, body background still `rgb(244, 238, 227)` in light scheme, no console errors. **To WP-06 (fonts):** `@ibasho/ui/fonts.css` compiles to 848 `@font-face` rules (714 kB CSS); the landing loads it after first paint to keep AC-L1, but a Latin-first subset entry (for example `./fonts-latin.css`) would help both apps. Small woff slices also get inlined as data URIs by Vite's default `assetsInlineLimit`; the landing turns that off for fonts.

**builder:WP-12** interface · open
**Request to WP-05 (companion mount, one line plus discovery):** add `@ibasho/agent-bridge` to `apps/companion` dependencies at the next install window, then `app.route(BRIDGE_MOUNT, createBridgeApp({ hub, token }))` with `token = createServerToken()` and `hub = new AgentHub({ onLedger })`; bind `BRIDGE_HOST` (127.0.0.1); on listen

`writeServerJson({ port, token, pid: process.pid, startedAt, bridge: BRIDGE_MOUNT })` and print `hub.pairingCode().code`; `removeServerJson()` on clean exit. All exported from `@ibasho/agent-bridge/server` (constants from `@ibasho/agent-bridge`); snippet in `packages/agent-bridge/README.md`. The SSE transport needs no extra dependency; `/link/ws` turns on if you pass `upgradeWebSocket` from `@hono/node-ws`.

**builder:WP-12** interface · open

**Request to WP-08 (Settings > Agent link):** add `@ibasho/agent-bridge` to `apps/web` at the next install window. Toggle off by default with one line saying a connected agent may send what it reads to its model provider (SPEC 13); a code field; then `connectAgentLink({ origin: 'http://127.0.0.1:7357', pair: { code }, client: { app: 'ibasho-web', version }, handlers, onStatus, onLedger })`. `handlers` implements `LiveHandlers` (`packages/agent-bridge/src/protocol.ts`): `apply` runs ops through the OpValidator and the undo stack with an "agent" mark; `say` runs the narrator validator; `screenshot` returns a base64 PNG of the stage; `thread` draws the red thread. Call `link.pushState(state)` on route, chapter, layer and selection changes; keep `link.session` in sessionStorage to reconnect after a reload (`pair: { session }`); write every `onLedger` entry to the outbound ledger (SPEC 5.14). Every incoming call is already schema-checked in the client.

**builder:WP-04** interface · proposed

**To WP-02 (engine emitter), from WP-04:** the certificate now has optional slots the checkers need (`@ibasho/spec-gen cert.ts`): `groups[].optionMeta` (`{ [optionId]: { measure, kind } }`), `context.home` (`{ region, facts: homeFacts(twin), checks: twin.checks }`), `claimed.saving` (int64 [row][fuel] annual Wh, before minus after) and `claimed.carbonAnnual` (int64 [row] g, `carbonSavingG`). Until the engine emits them, `@ibasho/verify` `emit(result, { optionMeta, home })` fills them (saving and carbon from the certificate physics). Please emit them from `emitCertificate` so the checks test the engine's own claims. Also: `SPEC_HASH` is now real (`ab8d651f...`), so the result hash changed; run `pnpm --filter @ibasho/engine gen:results` to refresh `packages/fixtures/src/results/gh1-m0.search.json` (engine `results.test.ts` and mcp `l2-engine.test.ts` are red until then).

**reviewer:WP-04** interface · open

**To WP-02 and WP-04 (M-P): `context.memory.completed`.** SPEC AC-P1 says a measure marked done leaves future searches. Today the engine copies `memory.completed` into the certificate but does not read it, and neither `ibasho-check` nor the mirror checks it, so a certificate offering a completed measure passes. Proposal: once WP-02 defines the semantics (exclude like `rejected`, or fold into the twin), WP-04 adds it to `deriveRow` in Lean and the mirror, with a mutant.

**builder:WP-04** interface · proposed

**To WP-02 (engine emitter), checker gaps closed:** the checkers now test three more claims. (1) `money.bills`: `claimed.bills`, int64 [row][fuel] annual your-home bill in micro-pence (slab-exact, `billOf` of the row's delivered energy, standing charge rule as in your certificate entry). Missing means the check fails, so `@ibasho/verify` `emit` fills it from the certificate physics until `emitCertificate` does. (2) `score.recompute`: optional `claimed.score = { zone: 'uk' | 'in', mode: 'burden' | 'per-area', rows: [{ row, indicators, sub, groups, total }] }`, rationals as `Q.toWire` strings, rows strictly increasing, and every front and pick row present; sub, groups and total must equal `subScore`, `groupScore`, `score` of the indicators exactly. Without it the check is skipped with its reason. `emit(result, { ..., score: { zone, mode, indicators } })` and `scoreClaim(...)` build the block from indicators until the engine emits its own. (3) `context.memory.completed` (SPEC AC-P1): an option id, or an option of a measure id, in `completed` is excluded exactly like `rejected` (status blocked, `BUILTIN.MEMORY`). The engine copies `completed` but does not exclude it yet, so a real search with a done measure now fails `feas.recompute`; please treat `completed` like `rejected` in search. This answers the reviewer entry on `memory.completed`.

**steward:WP-00** interface · final

**Bad outcomes contracts (SPEC 14), additive, in `@ibasho/core` (`packages/core/src/maintenance.ts`):**

- **`maintenance` plugin kind** (engine, numeric, so provenance with sources and grade is required). `MaintenancePlugin`: `ages` (`{ kind: 'measure', measureId }` or `{ kind: 'system', category, systemIds? }`), `neglect` (message key), `effects: DecayEffect[]`, optional `endOfLife: { ageYears: { lo, mid, hi }, replaceWithSystemId?, explain }`. `DecayEffect`: `param` (JSON Pointer into the twin, as in `TwinPatch`), `unit`, `rate` (`{ form: 'relative', ppbPerYear: { lo, mid, hi } }` with bigint ppb, same recurrence as discounting,

negative means the value falls; or `{ form: 'absolute', perYear: { lo, mid, hi } }` in `Q`), `onsetYears?`, `onsetFrom?: 'install' | 'as-of'`, `clamp?: { min?: Q, max?: Q }`, `quantity` (the evidence quantity id of the aged value), `explain`. Registry check `maintenance` (`checkMaintenance`): ordered ranges, pointer params, a target, effects or an end of life. WP-01: decay models go in `packs/home-uk/src/engine/maintenance/`.

- **"If nothing is done" run:** optional `EngineApi.doNothing?(req: DoNothingRequest): Promise<DoNothingResult>`. Request: `twin`, `owner`, `asOf`, `mode?` (default `your-home`), `horizonYears?` (default `owner.horizonYears`), `reportYears?`, `rate?: 'lo' | 'mid' | 'hi'`. Result: `hash`, `specHash`, `maintenanceFingerprint`, `mode`, `horizonYears`, `years: { year, deliveredWh: per fuel, bill, carbon, score? }[]`, `effects: { maintenanceId, param, year, evidence, band? }[]`, `replacements: { maintenanceId, dueYear: { lo, mid, hi }, cost?, evidence }[]`, `totals: { costNpv, carbon }`, `negatives`, `notApplicable` (model ids that matched nothing), `graph`. Every value is an evidence id, never a raw number (AC-B1). WP-02 implements, WP-08 draws the fourth line, WP-12 may expose it.
- **Negatives:** `Negative { indicator, kind: 'fuel-rise' | 'money' | 'carbon' | 'performance' | 'disruption' | 'risk', evidence, before?, after?, risk?: 'moisture' | 'overheating' | 'noise' | 'fire' | 'structural' | 'other', ruleId?, maintenanceId?, message }`; `NegativesByPackage = Record<packageId, Negative[]>`. `SearchResult.negatives?` covers baseline, front and picks (like `graph`); optional `EngineApi.negatives?(resultHash, packageIds)` for the rest. AC-B3 reads these. WP-09 writes the `message` copy.
- **WP-02 request (3), now applied:** `OwnerCtx.escalation?: { scenario: 'low' | 'central' | 'high', ppbPerYear?: per fuel }` (`EscalationScenario`, `ESCALATION_SCENARIOS`; scenario values belong to the money pack) and `OwnerCtx.counterfactual?: { replacementDue: Tri, systemId?, cost? }` (A_cf counts only on `yes`, DEC SC-F5). Absent fields keep M-0 behaviour. `SystemCategory` is now a named type.
- New `EngineApi` members are optional so current implementers still compile; they become required at M-B. `PLUGIN_KINDS` has 32 entries.

**builder:WP-07** interface · final

**Scene entry for the app shell (WP-08).** `import { HouseStage } from '@ibasho/scene'`; `<HouseStage twin={twin} quantities={q} style="washi" />` fills its parent (give it a sized box). `style`: `washi`, `yoru` or `svg` (no WebGL). Optional: `result` (quantities then come from `result.graph`), `layers` (default `heat-flow`, `zone-tint`), `pencil` (underdrawing while computing), `reducedMotion`, `tierCap`, `llmGenerating`, `store` (shared selection), `showOutline`, `assetsBase` (default `/app/assets/`, `null` stays procedural) and `stage` (any other `Stage` prop). This replaces `apps/web/src/home/HouseDrawing.tsx` at M-A. **Pack ui (`@ibasho/pack-home-uk/ui`):** `homeUkUiModule` (default export of `src/ui/module.ts`, id `pack-home-uk/ui`) loads 16 `measure-ui` plugins, ids `home-uk.measure-ui.<measureId>`; `mon` values are keys of `MONS` in `packages/ui/src/mon/set.ts` (WP-06: please export `MONS` or `isMonId` from `@ibasho/ui` so a contract test can check them). `FoldSpec.targets` are element kinds plus `system` for plant drawn as props. `ArchetypeThumb` and `thumbnailDrawing(form)` draw every `BuiltForm` as an ink elevation (WP-08 Shape step can use them instead of local thumbs). WP-10: the ARCH 4.3 contract "every measure id has a `measure-ui`" can import `HOME_UK_MEASURE_UI`.

**builder:WP-02** interface · final

**M-B engine outputs (WP-08, WP-09, WP-10, WP-12):** `EngineApi.flips(hash, param)` returns every crossing (`at` is p/kWh to 0.001 or percent to 0.001; a pick that empties reads `none`); `doNothing(req)` and `negatives(hash, ids)` are implemented. `SearchResult.negatives` covers the baseline, front and picks. New evidence quantities (your-home unless noted): `acct.saving.<fuel>` (annual), `money.bill_saving.<fuel>`, `fin.payback_simple` (exact years or cat `immediate`/`never`), `fin.payback_discounted`, `fin.irr` (exact percent or cat `n/a:<reason>`), `fin.loan_monthly`, `fin.cash_positive_m1`, `risk.<kind>.<ruleId>`; standard: `score.sub.<F..T>`, `score.group.<g>`, `score.total` (exact, or an interval while T is unknown), `score.delta`; rating: `epc.sap_est`, `epc.band_est` (cat `C (B to D)`). Do-nothing quantities: `<effect.quantity>.<modelId>.y<t>` (with `.lo`/`.hi`), `donothing.delivered.<fuel>.y<t>`, `donothing.bill.y<t>`, `donothing.carbon.y<t>`, `donothing.replacement_due.<id>`, `donothing.replacement_cost.<id>`, `donothing.cost_npv`, `donothing.carbon_total`. Negative message keys for WP-09: `negative.fuel-rise.<fuel>`, `negative.bill-rise.<fuel>`, `negative.year1-cost`, `negative.npv`, `negative.no-payback`, `negative.carbon-rise`, `negative.lifetime-carbon-rise`, `negative.score-fall`, `negative.band-fall`, `negative.risk.<kind>`, `negative.donothing.{fuel-rise.<fuel>,bill-rise,carbon-rise,replacement-due}`, plus each decay effect's own `explain` key. The `epc-band-at-least` target now works (central band).

**builder:WP-02** interface · open

**Requests from WP-02 (non-blocking):** (1) WP-04: spec-gen writes integer rationals as `"4/1"`, which core `Q.fromWire` rejects; the engine parses them with `specQ`. Either emit `"4"` or say this is intended. Also export

prices/reference-v1 (SAP 10.2 Table 12 prices, standing charges, primary energy factors) so packages/engine/src/score/reference.ts can go. The certificate does not yet carry A_cf or escalation; both are absent unless the owner sets them, so M-0 certificates are unchanged. (2) WP-00: EpcProxy.sap(run) needs the floor area (money-uk adds sapFor(run, tfaM2)); a money-pack kind for escalation scenarios (values now SF 5.1 defaults in finance/owner.ts); a disruption field on MeasureModel (days, rooms) for disruption negatives. (3) WP-01: put PV yield and self use on SimResult (the R indicator uses exported energy as a lower bound); write the decay models in packs/home-uk/src/engine/maintenance/ (the engine runs any maintenance plugin now; tests use a boiler model of −1% a year, end of life 12, 15, 20 years).

**builder:WP-03** interface · final

**Rules M-B for WP-02, WP-04, WP-08, WP-12** (all from @ibasho/engine, source packages/engine/src/rules). evaluateRules now uses DEFAULT_FNS (pass fns: {} for the old all-unknown behaviour), merges K_FACTS (k.*) into every env and derivePackageFacts(options, home) under ctx.pkg for package rules (p.kind.<kind>, p.adf.{major,minor,now}, p.vent.provided, p.airtighter, p.export.{W,limitW}, p.envelope.share, p.funded.pas2035); packageEnv(ctx) returns the exact package env. homeFacts covers RP 4 (h.wall.u, h.wall.solidMasonry, h.art4.P14.A style codes, h.dhw.class, h.vent.ap50, h.floor.type, h.envelope.areaM2, h.past.adf.*, ...). packageFacts takes fundedPas2035?. Built-ins add BUILTIN.TENURE.{TENANT,LEASE}, BUILTIN.PREREQ.BATTERY-PV, BUILTIN.CONFLICT.{EWI-IWI,GLAZING}. Also: normaliseObligations, obligationPatch(twin, obligations) ({ patch, unpatched }, idempotent), renderRule/renderExpr, lintRules, noise({ twin, units }), noiseFact(twin, units), mcs020aLpLe, adfCategory, meesCapMp, dtHosting, ensRetv.

- **WP-02:** (1) wire EngineApi.noise in src/api.ts to noise from ./rules/index.ts (it returns NoiseResult synchronously). (2) Apply obligations with obligationPatch before re-running physics; map { kind: 'assessment', check: 'moisture-assessment' | 'pas2035-assessment' } to the costed options assess-moisture and assess-pas2035; emitter-flow and hp-meets-design-load come back in unpatched. (3) Pass fundedPas2035: true to packageFacts only when a stacked grant requires PAS 2035. (4) The rule pack grew, so regenerate gh1-m0.search.json (statuses are unchanged; outcomes and consents grow). (5) BUILTIN.MEMORY and BUILTIN.TARGET stay in the search.
- **WP-04 and WP-02 (certificate parity):** package rules now read derived p.* facts, and APPL.*/built-in blocks depend on them. The certificate should carry packageEnv(ctx) (or the derived facts) per row so feas.recompute sees the same env. Named functions are used only in consent/oblige/warn/info rules (lint fn-status), so status never depends on NAMED_FN_IMPLEMENTED. Lean mirrors to add when convenient: mees_cap, dt_hosting, ens_retv (TS is exact rationals; contracts in fns/*.ts headers).
- **WP-01:** emit the option facts listed in OPTION_FACTS (@ibasho/pack-rules-uk/engine), above all m.ashp.noise = noiseFact(twin, units), m.ashp.{volumeM3,units,mount,frontsHighway,nearerHighway,coolingOnly}, m.wall.share, m.ewi.{finish,matches}, m.pv.{mount,exportW,frontsHighway,projectionM}, m.battery.exportW. Until then those rules read unknown and list their consent (worst case), which is honest.
- **WP-09:** message keys used by rules are rules.<rule id lowercased>.explain plus each effect's reason/risk/note key; renderRule gives a fallback English sentence.

**agent-A:WP-00** interface · final

**V-0 contracts v3: what B and C can import now** (typecheck green; all additive, every v2 type unchanged).

- **@ibasho/core** (barrel re-exports packages/core/src/v3/**): building BuildingDoc, Storey, Room2D, Segment, WindowRule, ShadeRule, Opening, Room3D, Core, RoofSpec, FreeShade, ThermalZone, MassingSpec, TenureSplit, Mm, Mdeg, Mm2, PointMm, PolygonMm, BuildingTypeId/BUILDING_TYPES, FacadeBin/FACADE_BINS/facadeBin(mdeg); compiled CompiledBuilding, ZoneTwin, Quantities, Adjacency, PlanCheck, SurfaceSet (triangulated mm surfaces with ids for scene and site), isZoneTwin. Place Place, ContextBuilding (height lo/likely/hi mm plus source), ContextTree, CityTile (ibasho.city/1), CityPackPlugin, tileKey. Site SiteResult, Epw, Resolution, RESOLUTION_LEVELS, DaylightTier, SunHours. Results BuildingResult, EndUse/END_USES (normative order, PV negative), TariffBill, WaterBalance, ComfortHours, CodeRow, RatingRow, BaselineRow, SplitShare, Applicability/APPLICABILITY, euiIsPartial. Stages StageId, STUDIO_STAGES, STAGE_META (label and doing message keys, after), stageIndex, downstreamStages(changed), Tier/TIERS, ResultStamp and isPreview(stamp) (drives the Preview stamp, R26), StageIn, StageOut, CodesOut, StageEvent, AnyStageEvent, StageProvider, AnyStageProvider, JobCtx, JobRequest, JobEnd, StageStatus, StageBarState, StudioApi, SiteContext. Optimise OptimiseResult, Candidate, Sweep, dominates. Codes CodeEdition, editionFor, RatingStep, ratingStepFor. Goals and settings Goal, ScopedGoal, Objective, DEFAULT_OBJECTIVES, MemoryScope, resolveScoped, Setting, SettingDef, SettingId, SETTING_GROUPS, resolveSetting, affectsMatch, staleSet (H8). Partner DesignOp, RecSet, RecItem, RecRole,

ChangeExplanation, Proposal, DecisionRecord. Lens LensId, RampId, LensProps, LensLegendProps, AnalysisLensPlugin, EditorToolPlugin, EditorDepth, PlanPointer. Metrics MetricId, METRICS, BUILDING_FACTS, ZONE_FACTS. Units add kWh/m2.yr, W/m2, Wh/m2, lux, h/day, h/yr, L, kL, mm, mm2, mdeg, kVA, INR/yr, micro-paise/*. SealLevel and provenance kind gain preview. 15 new plugin kinds (ARCH 15.3) in PLUGIN_KINDS with registry checks.

- **@ibasho/core/schemas**: BuildingDocSchema, CompiledBuildingSchema, PlaceSchema, CityTileSchema, ContextBuildingSchema, SiteResultSchema, BuildingResultSchema, CodesOutSchema, OptimiseResultSchema, ResultStampSchema, EvidenceGraphSchema, plus toWireJson(value) (bigints as decimal strings; every schema accepts that wire form and returns bigints, so Schema.parse(JSON.parse(toWireJson(x))) equals x). Use these when reading stored or fixture JSON.
- **Layers** (.dependency-cruiser.cjs): plan, site, preview are L1; interop L2; editor L3 with scene and pack ui halves, and every L3 may import @ibasho/plan pure helpers. preview alone may read packages/fixtures/src/{preview,buildings}.
- **Installed (exact, `npm view` checked):** clipper2-ts 2.0.1 (plan), immer 11.1.18 (@ibasho/store; WP-15 adds it to packages/editor when it creates that package), react-resizable-panels 4.14.0 (@ibasho/web), d3-scale 4.0.2, d3-shape 3.2.0 and their @types (@ibasho/ui). Skeletons with READMEs: packages/{plan,preview,site,interop}, packs/{types-global,types-in,benchmarks,codes-in,codes-global,ratings,city-hyderabad,city-london}. packages/editor is left to WP-15 (B).
- **Coming next from A (same session):** @ibasho/preview (seven Preview providers, previewEngineModule, runPreviewJob, previewBundle(buildingId)), @ibasho/plan helpers and @ibasho/fixtures/buildings/* (GB1 to GB4, GH1), @ibasho/pack-city-hyderabad tiles. A HANDOFF lists the exact API when each lands.

**agent-A:WP-14** interface · final

**Place search: find a building, floor or room by name (for B's search box and C's CityStage focus).** Contract in @ibasho/core: PlaceHit { kind: 'building' | 'storey' | 'room'; id; name; aliases: string[]; buildingId; buildingName; storeyId?; city (pack id, e.g. 'city-hyderabad'); tile?: [tx, ty]; centre: [x, y] (place frame mm, the pack frame for the samples); bbox: { minX, minY, maxX, maxY } mm; heightRange?: { lo, hi } mm (building height range, or storey floor to top); source: 'osm' | 'document' | 'merged'; contextIds? (OSM ids a document replaces: hide them when focusing); score 0 to 100; matched: string[] (names that matched, for highlighting) } and PlaceSearchOptions { limit? (10), kinds?, city?, buildingId?, minScore? (30) }. Implementation in @ibasho/plan (pure, offline, fast enough per keystroke): const index = buildPlaceIndex({ cities: [{ city: 'city-hyderabad', tiles }], buildings: [{ doc, city: 'city-hyderabad', contextIds: HERO_CONTEXT_IDS.gb1 }] }) then searchPlaces(index, query, opts?) => PlaceHit[] (ranked: exact, prefix, token and fuzzy (edit distance 1 or 2); case and diacritic insensitive; every query word must match). Tiles come from loadCityTiles() of @ibasho/pack-city-hyderabad/engine and @ibasho/pack-city-london/engine; documents from @ibasho/fixtures/buildings/<id>.building.json (or the store). Names indexed: OSM name, alt_name, short_name, official_name, name:en, old_name, loc_name, addr:* (now kept in the tiles as ContextBuilding.aliases and .addr), curated campus names (CURATED_ALIASES: Vindhya Bhavan, Himalaya, Nilgiri, KCIS for Kohli Research Block, OBH, NBH, Bakul, Parijaat, AIMSCS, ...), storey names plus "floor N", "level N", "LN", "Nth floor", ordinals, "basement" or "cellar", and room names plus their function (e.g. "A3 lab"). Examples that work (tests): "Vindhya", "vindya", "VINDH", "KCIS", "Himalaya", "Vindhya floor 2" (storey Second floor), "A3 lab" (rooms), "vindhya cellar". Also exported: normaliseName, editDistance, CURATED_ALIASES.

**builder:WP-07 (agent C)** interface · final

**<CityStage> for the studio (agent B mounts it).** import { CityStage, type CityStageProps, type CityStageHandle } from '@ibasho/scene'; it fills a sized parent. All props optional; with none it shows the Vindhya stand in on the stand in IIIT campus, 21 March 10:30. Source of truth: packages/scene/src/city/CityStage.tsx and src/city/types.ts.

- Data: place?: { lat, lon, tzHours?, name? }; tiles?: CityTile[]; roads?; hero?: BuildingDoc | null; compiled?: CompiledBuilding | null (hero drawn from meshes, lens ids exact); site?: SiteResult | null (radiation and facade sun hours per surface id); daylight?: DaylightGrid[] | null; lanterns?: LanternDatum[]; stamp?: ResultStamp | null. Rule: undefined = labelled scene estimate (Preview), null = not computed yet (nothing painted).
- View: level? / onLevelChange? (city | site | building | floor | room), storeyId?, roomId?, onSelect? (CityTarget); lens? / onLensChange? (none | shadows | sun-hours | radiation | daylight | lanterns); time?:

`{ date: 'YYYY-MM-DD', minutes }` / `onTimeChange?`; `azimuthDeg?` / `onAzimuthChange?`. Each is controlled when given.

- Moments: `sweep?: { dates, key }` + `onSweepEnd?` (1); `penumbra?` (default true) and `heightOverrides?: Record<contextId, mm>` closes a band (2); radiation wick (3, automatic); daylight drawer at Floor level (4); `phototropic?: { t, canopyTiltDeg?, finDepthM?, steps? }` (6); `style: 'yoru'` or lens `lanterns` (8).
- Find (owner request): `focus?: { kind: 'building' | 'storey' | 'room', buildingId, storeyId?, roomId?, bboxMm?: { minX, minY, maxX, maxY } } | null` with `onFocusChange?`, and `ref.current.flyTo(focus | null)`. Eases the camera (cut under reduced motion), outlines in ink by day or lights by night, dims the rest, and for a storey or room lifts the floors above away (cutaway). A hero id maps to Building, Floor or Room; a context id frames that neighbour; any other id uses `bboxMm` (map a `PlaceHit` bbox here); unknown with no bbox does nothing. Choosing a ladder level clears the focus. `resolveFocus()` is exported for the search UI.
- Shell: `style?: 'washi' | 'yoru' | 'svg'`, `reducedMotion?`, `tierCap?` (default 2), `llmGenerating?`, `controls?` (default true: ladder, lens chips, time strip, legend; set false and drive by props), `legend?`, `legendFor()` for an external legend, `stats?`.
- Requests to WP-00 (core): add `DaylightGrid` (per storey sDA and ASE point grid) and `LanternDatum` to `core/v3`; confirm the `BuildingDoc.rotationMdeg` convention used here: plan +y points to that bearing clockwise from true north. Agent A: `packs/city-hyderabad` tiles drop straight into `tiles`.

**builder:WP-08 (agent B)** interface · open

**To agent A (WP-13), design levers in the Preview energy stage.** `computeEnergy` calls `energyModel(compiled, clim, {})`, so the studio's what-if levers that are not geometry (glazing SHGC and U, roof absorptance, wall U, LPD scale, daylight controls, cooling set point, PV kWp) do not reach it; the optimiser already turns them through `EnergyParams`. Please read them from the energy input, for example `StageIn.energy.params?: EnergyParams` (or settings `design.*`), and export `EnergyParams`. Until then `apps/web/src/studio/preview/levers.ts` `adjustEnergy` scales the energy result with the same factors (PLACEHOLDER, labelled). Window share, shading and the cooling system already go through the document (`variantDoc`, `doc.systems`). Also: the studio's setting ids now follow `PREVIEW_SETTING_DEFS` (`money.tariff.elec.rate` in rupees per kWh, `carbon.grid.factor` in g per kWh).

**builder:WP-06 (agent B)** interface · final

**To agent C (WP-07): Andon controls for the city chrome** (design review city-V-P B3). `@ibasho/ui` now exports `Range` (kumiko track with step buttons, `onInput` while dragging, `onChange` on commit, optional `mark`), `SunScrubber` (24 hour track: `minutes`, `onChange`, `sunriseMin`, `sunsetMin`, `altitude(m)` for the ember arc; arrows 5 min, Page 1 h, Home and End sunrise and sunset; `children` for a play button) and `Picker` (listbox with type-ahead; `quiet` for bars), beside `ScaleLadder`, `ChoiceChips`, `AnalysisLegend`, `PreviewStamp`, `Stale`, `TierCaption`, `StageBar`. The studio mounts `CityStage` with `controls={false}` and `legend={false}` and drives level, storey, room, time, lens, site, stamp, style and `focus` by props (`apps/web/src/studio/niwa/Niwa.tsx`); compare thumbnails use `style="svg"`.

**agent-A:WP-00** interface · final

**QA round 1 fixes: contract additions and behaviour changes for B and C** (all additive; nothing removed or renamed).

- **Contracts:** `Room2D.aliases?` and `Storey.aliases?` (other names, functions only); `PlaceHit.kind` adds `'place'` (campus or site; `buildingId` is its first building, `buildingName` empty), `PlaceHit.source` adds `'place'`, `PlaceHit.storeyName?` on storey and room hits; `SiteContext.replacedIds?` (context ids the building replaces). Exhaustive switches over `PlaceHitKind` need a `place` case.
- **Search:** pass places too: `buildPlaceIndex({ cities, buildings, places })` with `places.json`. "IIIT", "IIITH", "campus" give the place; "Shodh", "Bodh", "Vrithi", "Block D" give part hits with id `gb2#<name>` (kind `building`, `buildingId: 'gb2'`, bbox of that block: focus by bbox); rooms answer to real names ("Auditorium", "Seminar Hall I", "SBI Bank and ATM", "IIIT Library", "C-STAR Incubator", "Centre for IT in Building Science", "Robotics Lab B3"). Do not gate on score 50: typos score about 48.
- **Preview runs:** pass `replacedIds: HERO_CONTEXT_IDS[buildingId]` in `JobInputs` (else the building's own footprint decides). Unknown-height neighbours no longer shade (they stay ghosts). A late older generation now ends `superseded` at once and never replaces the live job. Progress events carry the tier already on screen (`rough` until the rough partial, then `refined`), never a later one. Daylight rows exist only for regularly occupied rooms and have no `ecbcShare`. ECSBC row is now `ECSBC.EPI-RATIO` (metric `epi.ratio`, margin as a ratio; `unknown` when the range straddles 1). HT bills are HT-II(A) per kVAh with time of day and demand from a

load sum (`demandKva`); flats use LT-I categories (the 200 unit cliff reprices every unit) with a Gruha Jyothi setting. New settings: `money.tariff.elec.powerFactor`, `money.scheme.gruhaJyothi`, `comfort.dhw.solarShare`. EUI stays per conditioned area; evidence `eui.site.gfa` gives it per gross area.
- **Data:** GB1 is refitted to its OSM outline (rotation 41.7, footprint IoU at least 0.8, 93 named rooms, 28,325 m² GFA); GB2's Bodh and Block D sit on their OSM ways, Shodh and Vrithi at assumed spots clear of Kohli Research Block. Regenerated `@ibasho/fixtures/buildings/*` and `@ibasho/fixtures/preview/*`: room ids changed for GB1 and GB2.

**builder:WP-08 (agent B)** interface · open

**To agent C (WP-07):** during the studio screenshot pass `CityScene.tsx` threw `ReferenceError: useCallback is not defined` at runtime (line 286; typecheck was green at the time, so likely an import dropped mid-edit). The studio now wraps `CityStage` in an error guard and the router has an Andon error page, so panels keep working, but Look and Make need the scene. **QA round 1 folded in by B:** search passes `places`, groups Places, Buildings, Floors, Rooms with `storeyName`, Himalaya blocks (`gb2#…`) focus by `bboxMm` (passed in the URL as `bbox`), jobs pass `replacedIds`, `ECSBC.EPI-RATIO` has a title, the ledger shows demand charge, power factor, Gruha Jyothi and solar hot water share.

**builder:WP-07 (agent C)** interface · final

**`<CityStage>` update after the design QA round (supersedes the props list in my earlier entry where they differ).** Source: `packages/scene/src/city/CityStage.tsx`.
- Real data: `tiles` (pack `loadCityTiles()`), `hero` (GB1 doc), `compiled` (`compile(doc)` from `@ibasho/plan`), `hideContextIds` (pass `SiteContext.replacedIds` or the pack's `HERO_CONTEXT_IDS`). Neighbours with `unknown` height are drawn as a guess with a hatched range and never cast solid shadows.
- Find: `focus` / `onFocusChange` / `ref.flyTo`; `focusFromHit(placeHit)` maps a `PlaceHit` (kinds place, building, storey, room) to a focus; ids the stage has no geometry for frame their `bbox`.
- New: `onHeightEnter(contextId, mm)` (pinned neighbour chip with a floor stepper; store it and pass it back in `heightOverrides`), `insets` (CSS px the studio's own chrome covers, so the camera frames around it). Exports `layoutOf`, `chromeInsets`, `radiationDomain`, `SWEEP_DATES`, `passMask`.
- Chrome: DS 13.7 layouts built in (phone bottom sheet at 55%, tablet breadcrumb and right legend, desktop vertical ladder); uses `ScaleLadder`, `SunScrubber`, `Stepper`, `PreviewStamp` from `@ibasho/ui`.
- Requests to WP-06: `AnalysisLegend` numeric ends without evidence ids, and the scene ramps `kage` (shadow hours) and `hinode` (sweep build up) in `RAMPS`, so the stage can drop its legend markup copy.

**agent-A:WP-13** interface · final

**To agent B (WP-08): design levers now reach the Preview energy stage** (answers the WP-08 levers request; additive, old callers unchanged). Contract in `@ibasho/core`: `EnergyParams { wwrScale?: Partial<Record<FacadeBin, number>>; shgc?: number; uWin?: number; uWall?: number; uRoof?: number; roofAbsorptance?: number; pf?: Partial<Record<FacadeBin, number>>; lpdScale?: number; setpointC?: number; daylightControls?: boolean; pvKwp?: number; dhwSolarShare?: number }` (all optional; absent is the building as drawn: SHGC 0.62, window U 5.6, wall U 1.9, roof U 1.8, roof absorptance 0.75, set point 24) and `StageIn.energy.params?: EnergyParams`. Pass it with `runPreviewJob(req, { ...inputs, energyParams }, ...)` or `createPreviewStudio().runJob(...)` (`JobInputs.energyParams`); the energy inputs hash includes it, so a lever change reruns energy, codes and optimise. Presets from `@ibasho/preview`: `PARAM_PRESETS.glazing['low-e']` (SHGC 0.3, U 2.8), `.roof.cool` (absorptance 0.3), `.wall.insulated` (U 0.4), `.lighting.led` (LPD x 0.65), `.daylight.on`; `pvKwpFor(compiled, roofShare)` gives kWp at 6.5 m² per kWp. Window share, shading and the cooling system stay in the document. With this `apps/web/src/studio/preview/levers.ts` (PLACEHOLDER) can go. Also new: `SiteResult.unknownHeightIds?: string[]` lists neighbours inside the radius left out as shade because their height is unknown (GB1: 29), so a lens can say "29 neighbours of unknown height ignored; shade may be understated".

**agent-A:WP-17+WP-13** interface · final

**On-demand GCC offices (Qualcomm, Vanguard), one canonical bill, faster Preview.**
- **New on-demand area** `city-hyderabad-raidurg` (Raidurg and Knowledge City, HITEC City; 44 OSM buildings, about 6 KB): exported from `@ibasho/pack-city-hyderabad/engine` as `cityHyderabadRaidurg` (city-pack plugin, registered by the module but its tiles are only fetched on request), `loadRaidurgTiles(read?)`,

RAIDURG_TILE_FILES, and CITY_AREAS keyed by pack id: { onDemand: boolean; loadTiles(read?) } (city-hyderabad false, city-hyderabad-raidurg true). The area has its own frame (origin 17.4300 N, 78.3795 E). HERO_CONTEXT_IDS gains gb5: ['osm_w1133460476'], gb6: ['osm_w1133269186'].

- **Contract:** Place.onDemand?: true (schema too). Do not list or load such places by default; show them when search returns them or the person picks them, then load CITY_AREAS[place.cityPackId].loadTiles().
- **Place and buildings:** pl_knowledge_city ("Raidurg and Knowledge City", onDemand: true, cityPackId: 'city-hyderabad-raidurg', buildings gb5, gb6). gb5 Commerzone Raidurg (host of the Qualcomm Hyderabad campus; L-shaped twin tower, ground plus 19 floors, 3 basements, two service cores, 280,000 m2 GFA), gb6 Sattva Knowledge Park Tower 1 (assumed host of the Vanguard Global Value Center; octagonal plan, ground plus 23 floors, one core). Footprints are the OSM outlines (tested IoU at least 0.8, no overlap with neighbours); no public floor plans were found, so plates, storeys and glazing are estimated (see gb5.notes.json, gb6.notes.json, sources and unknowns). Preview fixtures in @ibasho/fixtures/preview/gb5 and gb6. Search: "Qualcomm", "QCOM", "Commerzone", "Vanguard", "Sattva Knowledge Park", "Knowledge City", "Raidurg" (pass pl_knowledge_city in places and the area tiles in cities to buildPlaceIndex, e.g. lazily once a query matches).
- **One canonical bill (A-M3):** headlineBill(result) from @ibasho/preview gives { total, currency, whole, byCarrier, inputsHash }; read every displayed bill through it from the latest energy or codes result. Same inputs give the same bill; it changes only with the document, settings money.tariff.elec.rate, .demand, .powerFactor, money.scheme.gruhaJyothi, comfort.setpoint.cool, comfort.dhw.solarShare, JobInputs.energyParams, whether radiation ran in the job, and the place. StageIn.optimise gains radiation? and params? (the runner passes them), so the optimiser's no-lever option reproduces the energy bill exactly (test).
- **Speed (A-M2):** replay defaults to DEFAULT_DEVICE_FACTOR 0.1 (pass 1 for real timings, 0 for instant); compiled geometry and the energy model are cached per document and inputs, so tiers share one compute. Test: each rough stage within 300 ms on GB1 (compute only).

**builder:WP-07 (agent C)** interface · final

**CityStage additions: studio review and Design before you build (SPEC 27).** Source of truth: packages/scene/src/city/CityStage.tsx, placement.ts, CityCompare.tsx.

- Night: while the stage is dark (Yoru style, Lanterns, or sun below the horizon) a sun lens (Radiation, Sun hours, Shadows) shows as Lanterns; the chosen lens is kept and returns by day, so B need not switch it. The hero is always the brightest lantern; other lanterns are capped below it.
- Penumbra: hatched only for up to 10 estimated neighbours whose shadow can reach the hero (preferring SiteResult.unknownHeightIds when site is passed), plus the pinned or hovered one; the rest show dashed tops; none at City level.
- Framing: Building and Floor levels fit the projected box of the building (height included) inside the chrome safe area with a 12% margin. Pass insets for B's own chrome.
- Orientation: for a north up 3D beside a north up plan, pass azimuthDeg={180}; the compass shows north at every azimuth.
- Room labels (one label rule): function first, then the room's own name; shown only when they fit inside the room on screen (else function only, else hidden), collision culled.
- Placing: placement?: { origin: { x, y } (place frame mm), rotationMdeg, groundMm? } (controlled) with onPlacementChange(placement); placeMode lets the person drag the building and turn it by a blue handle, with button steps (move 5 m, turn 15 degrees, ground 0.5 m); onSiteCheck(check) gives { overlaps: contextId[], shading: { id, name?, hours, heightM, estimated }[], free } (21 Dec and 21 Mar, 08:00 to 16:00; unknown heights never shade). Helpers: applyPlacement, movePlacement, placementCheck, placementOf.
- Compare: <CityCompare base={stageProps} options={[{ id, label, placement?, overrides? }]} level? lens? time? onSiteCheck? /> shows two or three panes with synced scale, lens, time and azimuth (one shared bar); overrides takes another city (tiles, place, hideContextIds) for London against Hyderabad.

**agent-A:WP-17+WP-14** interface · final

**Lloyds Technology Centre (owner priority), on demand.** Host: Sattva Knowledge City, Octave Block, Raidurg (the company's public address is floor 10 there). New building gb7 "Sattva Knowledge City, Octave Block" in place pl_knowledge_city (now buildings gb7, gb5, gb6; onDemand: true, area city-hyderabad-raidurg). OpenStreetMap does not name the blocks, so the outline (way 852979202) is an assumed match; ground floor plus 17 office floors, 3 basements, three cores, about 186,000 m2 GFA, all estimated (sources and

unknowns in `gb7.notes.json`). The tenant floor is its own storey: `st_tenant` “Floor 10, Lloyds Technology Centre” (aliases Lloyds, LTC, Lloyds Banking Group). `HERO_CONTEXT_IDS.gb7 = ['osm_w852979202']`. Search: “Lloyds”, “Lloyds Technology Centre”, “LTC”, “Octave Block” return `gb7` (storey hit for floor 10 with `kinds: ['storey']`). Preview fixtures in `@ibasho/fixtures/preview/gb7`. Office tower room and storey labels are now customer-facing (“Office floor, centre, wing 1”, “Entrance lobby, edge 2”, “Basement levels (3)”), and notes read as plain sentences.

**agent-A:WP-13+WP-17** interface · final

**Design before you build (SPEC 27), platform part.** All additive.

- **London plots on demand:** `@ibasho/pack-city-london/engine` exports `cityLondonKingsCross`, `loadKingsCrossTiles(read?)`, `KINGS_CROSS_TILE_FILES` and `CITY_AREAS` (`city-london` default, `city-london-kings-cross` on demand; 611 buildings, 184 with tagged levels or heights, 694 trees, about 52 KB). Place `pl_kings_cross` (“King’s Cross, London”, `onDemand: true`, `codePackIds: ['codes-uk']`, no buildings yet: place designed buildings there; its own frame from 51.5352 N, 0.1250 W).
- **Codes by plot location:** `codePacksFor` gives `codes-uk` for UK places and `codes-in` for India when a place lists none. New buildings in England get `UK.PARTL.ENERGY` (design EUI against the Part L 2021 notional, ratio band), `UK.PARTL.FABRIC` (unknown until constructions are set), `UK.FHS.2025` (fails with gas heating; else ratio against the Future Homes notional with a heat pump), `UK.PARTO.GLAZING` (dwellings, simplified method, high risk in central London) and in Greater London `LONDON.PLAN.SI2` (35% on-site saving for major developments) and `LONDON.PLAN.BE-LEAN`. Baseline row `Part L 2021`. Existing UK homes (permit before 2022-06-15) keep `UK.EPC.ESTIMATE` only. Indian plots unchanged. `EnergyParams.heatPumpScop` (heat pump for heating and hot water) is new.
- **Typed targets with gaps (AC-PD3):** core `Goal` gains `{ kind: 'net-zero-ready'; basis: 'uk-nzcbs' | 'fhs-2025' | 'london-plan' | 'in-nzeb' }` and `{ kind: 'lowest-opex' }`; `GoalGap { presetId?, goal, status: 'met' | 'not-met' | 'unknown', value?, target?, unit?, ruleIds?, reason, sources, grade }`; `GOAL_PRESETS` (ids `net-zero-ready-uk`, `fhs-2025`, `part-l`, `london-plan`, `ecsbc-super`, `leed-gold`, `lowest-opex`, each with sources and regions). Pass `JobInputs.goals`; the codes stage returns `CodesOut.gaps` (schema updated). `lowest-opex` needs the optimiser: call `goalGaps(goals, { codes, ratings, baselines, energy, optimise })` from `@ibasho/preview` after the optimise stage. LEED levels stay `unknown` (only the energy credit is estimated); ECSBC tiers read the EPI ratio (Compliant 1, Plus 0.85, Super 0.7, estimates).
- **Placement (AC-PD2):** `placementChange(prev, next)` gives `none`, `move`, `rotate` or `design`; `stagesForPlacementChange(kind)` the stages to rerun. A move reuses the compiled geometry (only placement and hash change, no recompile), so the geometry stage is instant and the sun onward reruns with the new context; pass the previous outputs as `JobInputs.previous`.

## Blockers

**builder:WP-12** blocker · open

Update to the WP-12 blocker: L2 is no longer blocked (green against the real engine). Still waiting: L5 live e2e (WP-05 mount, WP-08 Settings and handlers), `ibasho_verify` mirror (WP-04), L8 evals need a run of Claude Code against the plugin.

**builder:WP-02** blocker · open

WP-01 physics edits in progress break two engine tests: `light path differs from full simulation for heat-ashp+loft-300 in rating` (the heavy and light split no longer matches a full run) and fixture drift. With committed physics the engine suite passes (136 tests). WP-01: when the physics lands, check the factorisation test and run `pnpm --filter @ibasho/engine gen:results` (or ask WP-02).

**agent-A:WP-00** blocker · open

**To WP-03 (or the orchestrator): one-line fix in `packages/engine/src/rules/render.ts`.** The v3 `comply` effect (`{ kind: 'comply'; metric; op; limit; tier? }`, ARCH 15.4) makes `renderEffect` non-exhaustive, so `@ibasho/engine` typecheck fails (TS2366 at line 247) and the error shows in every package that type-checks engine sources (apps/web, packs/rules-uk). Add a `case 'comply'` that returns a sentence such as “metric must be op limit”. Not my path, so not edited.

**agent-A:WP-00** blocker · resolved
The engine `render.ts` `comply` case blocker is resolved: a `comply` case is now in `renderEffect` (added by another agent).

## Handoffs

**steward:WP-00** handoff · done
**SPEC 14 contracts landed** in `@ibasho/core`: `maintenance` kind and `MaintenancePlugin`, `DoNothingRequest`/`DoNothingResult`, `Negative`/`NegativesByPackage`, `SearchResult.negatives?`, optional `EngineApi.doNothing?` and `negatives?`, `OwnerCtx.escalation?` and `counterfactual?`. See the Interfaces entry. Core tests 140 pass; typecheck of core and its 22 dependants passes. Not done here: decay models (WP-01), the run itself (WP-02), the fourth line (WP-08), no zod schema for `DoNothingRequest` yet (add when a wire boundary such as MCP needs it).

**builder:WP-07** handoff · done
**Scene, M-A slice.** What exists: the stage (Washi and Yoru still life, SVG fallback and pencil), procedural house from the twin, heat-flow motes, thermal lens, zone tints, roof lift, picking, `a11y-scene` outline, quality ladder, and now `HouseStage` (see Interfaces) and the `home-uk` ui half (`measure-ui` for all 16 measure ids, archetype thumbnails for all 9 built forms). Look fixes: gables are plaster on the inside, so the far wall of the cutaway reads as a room; the stage paper grain is lighter (16 fibres, lower opacity). Tests: `@ibasho/scene` 10 files, 59 passed (new: cutaway composition, house build invariants, `HouseStage` defaults); `@ibasho/pack-home-uk` 4 files, 30 passed (new `test/ui`, 7). Typecheck green on both; Biome, dependency-cruiser and `tools/lint` clean on owned paths. Not built yet: M-B layers (ghost, noise rings, fold assembly, split, miniatures), people, `packs/home-in/src/ui` (T3). G7 frame rate on Iris Xe still to be recorded.

**builder:WP-08** handoff · done
**M-A shell, onboarding, Reveal.** `/app/start` runs five typed steps (`apps/web/src/start/steps.ts`: time budget guard, refuses any money question, R24). Offline place list and local postcode area read. Every unanswered value is an estimate and marked (AC-G2). `/app/` goes to the last own home, else `/start`. Reveal navigates to `/app/home/:id` with `state.reveal`; `apps/web/src/home/HomeHouse.tsx` mounts `@ibasho/scene` `HouseStage` (washi or yoru from the lantern, pencil while computing, `ROOF_LIFT` once when ready, skipped under reduced motion); no WebGL or a stage error falls back to the SVG house, pencil then ink. Layout zones are one component, `apps/web/src/shell/Zones.tsx` (Niwa, Engawa, Zashiki, Tokonoma). `manifest.ui` now loads `@ibasho/pack-home-uk/ui`; all packs by package name. Stubs: `start/seed.ts` seeds from the GH1 survey twin with every value re-tagged as an estimate until WP-01 `ArchetypePlugin.seed()` exists (only the terrace and Not sure tiles are pickable); all places use London weather; Tune order is fixed until `valueOfInformation`. Tests: `apps/web` 14 passed (3 files, incl. a seeded home reaching the ready state on the real engine worker body), typecheck and Biome clean. For WP-10: DOM hooks `.ib-home[data-state]`, `.ib-house[data-renderer]`, `.ib-choices[data-estimated]`.

**builder:WP-06** handoff · done
**UI kit, M-A and M-B sets.** `import '@ibasho/ui/andon.css'` (now also pulls `kit.css`). M-A: `Sheet` (side sheet from 900 px, bottom sheet with peek, half, full; every snap has a button; Esc closes; not modal), `NarratorStrip` (sentences settle one by one; “Plain narrator” marker; chips without seal glyphs, R11; `sentencesFromRendered(RenderedNarration)` adapts WP-05 output), `ScoreGroups` (numeral 0 to 100, four group bars, seven parts on tap, band with likely dot and “Sharpen” while material fields are unknown, the typical household line, no letter), `EpcChip` (“Estimated EPC”, spread either side until `bandValidated`). M-B: `Futures` (fixed order, four figures plus “More”, `Negatives` on every card even when empty, withheld and same-as states, and `DoNothingLine` “If nothing is done” with wear effects, decay sources, grade badges, replacements, negatives and models that do not apply), `MeasureCard` (six states with pattern and label), `FuelBars` (diverging, hatch for increases, range band, table toggle), `Slider` (kumiko bar, discrete evidence stops, keys and step buttons), `WhyNot` (rule, memory, tenure, witness, target, budget), `CheckLane` (names the check, cost with `GradeBadge`, record done), `MonthRing` (twelve wedges stacked per fuel with hatches, arrow keys wrap, tap or drag, table), plus `ChartFrame`/`TableAlt`, `GradeBadge`, `Mon` and `MONS`/`isMonId`/`monFor` (WP-07 ask), `magnitude`. All props carry `EvidenceId`s; every number is a `Num`. Adapters: `negativeItems(core Negative[], text)` maps WP-02 negatives.

Stories under `Kit/*` (`stories/kit-evidence.ts` is illustrative only). Not built: the rest of DS 8 (M-D), a Latin-first fonts entry (WP-09 ask, next), axe runs over stories.

**builder:WP-02** handoff · done

**M-B pieces without new physics. What exists:** `packages/engine/src/finance/{returns,owner}.ts` (simple and discounted paybacks, IRR only on one sign change by integer bisection, exact loan annuity and cash-positive flag, escalation scenarios, A_cf); `src/score/` (exact rational score, sub-scores and four groups from spec-gen constants, inputs from the standard run, estimated EPC band with one band either side); `src/flips/` (every crossing, B9 two-flip slab fixture passes); `src/search/{extras,do-nothing}.ts` and `src/evidence/outcomes.ts` (outcome nodes, per-package negatives, the "If nothing is done" run); `api.ts` implements `flips`, `doNothing`, `negatives`; the band target works. `packs/money-uk`: costs for every SPEC 9 option and the like-for-like boiler, loan stub, `epc-proxy/rating`. **Tests:** `packages/engine/test/mb.test.ts` (B4 weight and IRR injections, B5 dominance property, B9, AC-B1, AC-B3), engine 136 and money-uk 16 pass on committed physics. **Stubbed:** T needs 5R1C comfort (score is a range until then), renewable fraction from export only, reference prices in the engine, uncertainty, SEG, value range nodes, disruption negatives, real-scale fixture (needs WP-01 measures). See the WP-02 interface entries.

**builder:WP-03** handoff · done

**Rules, gate M-B (C1 to C7).** What exists: `packages/engine/src/rules/` adds `fns/` (`mcs020a_Lp_le` rational enclosure mirroring Lean `Noise`, `adf_category` mirroring Lean `Adf`, `mees_cap`, `dt_hosting`, `ens_retv`, `noise()` fast path from site geometry), `derive.ts`, full RP 4 `homeFacts`, `obligations.ts`, `render.ts`, `lint.ts`, built-ins (budget, tenure, prerequisite, conflicts). `packs/rules-uk` has 50 rules: `ENG.PD.*` (heat pump noise, volume, count, highway, listed, roof, cooling, retired 1 m boundary, Article 4, PV, EWI), `ENG.LB.ANY`, `ENG.BR.*` (renovation with 0.70 as trigger, flow 55, sizing, AD F categories, overheating advisory, traditional walls), `ENG.MOIST.IWI` (+ costed assessment), `UK.DNO.{HP,HP.LOOPED,G98,G99}`, `UK.PAS2035`, `UK.BUS`, `EW.MEES.{NOW,CAP}`, `APPL.*` (5), Wales and Scotland stubs; every rule graded and sourced, reviewer-memory rules cite DEC SC-3 or SC-4. Tests: engine rules 45 (C1 golden enclosure inside 31.95 to 32.05 dB, upper end open, C3, C4 adf monotone and obligations idempotent, C6 noise p95 under 5 ms, renderer, lint), rules-uk 72 (one golden per rule and built-in, the six SPEC 9 acceptance cases, C5 retirement on 2025-05-29, C7 lint and sentences, GH1 M-0). Stubbed: Wales and Scotland rule bodies, `rules-in` (T3). Needs others: see the WP-03 M-B interface entry (WP-02 `noise` wiring, obligations, fixture regeneration; WP-01 option facts; WP-04 certificate env).

**builder:WP-07 (agent C)** handoff · done

**City stage, V-P prototype.** What exists in `packages/scene/src/city/`: `CityStage` (one orthographic demand-rendered canvas, still when idle), merged context massing (one draw call) with ink or dashed outlines, instanced trees, BVH picking with a provenance chip, the penumbra and uncertain top bands, the sun from suncalc with day and year scrubbers and the 12 s sweep, a preview sun and shadow hours kernel (`shadow.ts`), the Hinata, Hidamari, Akari and Sora ramps, lens paint by shader injection (no extra draw calls), the radiation wick, the daylight drawer, the phototropic canopy and fins, Yoru lanterns, find and fly to (focus), the quality ladder, reduced motion and the `CitySvg` plan fallback. About 28 draw calls and 80k triangles at Site level on SwiftShader. Stubbed (PLACEHOLDER, labelled on screen): `standInCity()` until WP-17's pack, `vindhyaStandIn()` until the traced fixture, `previewSite()` and `previewDaylight()` until WP-13 and WP-16. Fixed: suncalc 2 returns degrees from north; `styles/sun.ts` (house light) now reads it correctly. Harness: `dev/city.html`; capture `name=city:<query>`. Tests: `test/city.test.ts` (25), scene suite green.

**agent-A:WP-00** handoff · done

**V-0 contracts, plus WP-13, WP-14, WP-17 prototype data (Agent A).** See the two Agent A interface entries for the full import list. Late additive contract changes: `CompiledBuilding.placement?` (origin and rotation, so site and scene place meshes), `ContextBuilding.holes?`, `.aliases?`, `.addr?`, `StageIn.energy|codes|optimise.site?` (place and weather), `StageIn.optimise.settings?`, `OptimiseResult.evidence?`, `PlaceHit` and `PlaceSearchOptions`. Ownership map now covers WP-13 to WP-24; depcruise has the v3 layers (preview may import plan and its fixture folders). **How B uses it:** manifest engine ref `{ id: 'preview/engine', load: () => import('@ibasho/preview').then((m) => ({ default: m.previewEngineModule })) }`; or call `createPreviewStudio({ deviceFactor: 1 }).runJob(req, { doc, place, tiles, settings, goals, objectives, previous }, onEvent, onEnd)` inside the engine worker; `cancel(jobId)`; events are `AnyStageEvent` with rough and refined `partial`s and one `final`; outputs

keyed by stage. `PREVIEW_SETTING_DEFS` seeds the settings ledger. Sample data: `@ibasho/fixtures/buildings/{gb1,gb2,gb3,gb4,gh1}.building.json` (parse with `BuildingDocSchema`), `places.json` (`pl_iiit` holds gb1 and gb2 with `cityPackId: 'city-hyderabad'`), recorded outputs in `@ibasho/fixtures/preview/<id>/<stage>.json` (parse with the result schemas). Edited documents recompute, so what-if and update flows respond. **How C uses it:** `compile(doc)` from `@ibasho/plan` gives `meshes.surfaces` (building frame mm, z up; ids `<segId>_wall`, `<segId>_win`, `<roomId>_roof`, `_floor`, `@k` per repeated storey) and `placement`; `toPlaceFrame(doc, p)` puts them in the pack frame. City: `loadCityTiles()` from `@ibasho/pack-city-hyderabad/engine` (tile URLs via `new URL(..., import.meta.url)`, gzip handled), hide `HERO_CONTEXT_IDS.gb1` / `.gb2` where the modelled building stands. Lenses: `SiteResult` from the sun (sun hours, svf), radiation (monthly Wh/m² per mesh surface id, window shading factors, ledger by OSM blocker id) and daylight (sDA, ASE, views per room) stages. **Data:** GB1 Vindhya 77 rooms over a cellar and three floors, 20,011 m² GFA, facades NE, SE, SW, NW; Preview EUI 176 to 243 kWh/m²·yr (data centre and labs), bill about 3.3 crore rupees a year; GB2 Himalaya, GB3 HITEC office (15 storeys), GB4 apartments (slab bills, cliff months), GH1 London. City Hyderabad 1,036 buildings, 68 trees, 72 KB. **Tests (each run once, capped):** core 146, plan 16, preview 21 (P1 to P4, cancel, H9), city-hyderabad 2, city-london 1, tools/city 3 (Y3), tools/lint 16; typecheck green for all owned packages; lint:tools clean; depcruise clean on owned packages. **Stubbed or open:** `compile` HomeTwin for GH1 (V-1), Open Buildings heights and terrain (V-2), skeleton packages site, interop, types, codes, ratings packs; `packages/editor` not created (WP-15); engine `render.ts` blocker (comply case).

**builder:WP-06+WP-08+WP-15 (agent B)** handoff · done

**V-P product UI: studio, five flows, editor, store v2. What exists.** `@ibasho/ui` v3 kit (see the WP-06 interface entry and `packages/ui/README.md`): StageBar, PreviewStamp, Stale, TierCaption, ApplicabilityChip, ScaleLadder, ModeTabs, LensPanel, AnalysisLegend, Histogram, OKLab ramps, TypeCards with eight type mons, EuiGauge, EndUseBar, RatingLadder, LedgerRow, Stepper, ChoiceChips, Range, SunScrubber, Picker, CompareGrid, FilmStrip, buttons; `Num` draws seal `preview` with a dashed underline and no glyph; stories `Studio/Kit (v3)`. New `@ibasho/editor` (WP-15): PlanEditor (9 tools, keys, touch, typed sizes, snap, underlay image), Inspector, StoreyStack, FacadePanel, PlanOutline, Immer patch commands with undo and redo, live checks and WWR and PF meters. `@ibasho/store` v2 (`V2`, `createStudioStore`, branch with lineage, visits). `apps/web` studio: `/` (places, Welcome back with what changed and old and new figures, Start over, Branch), `/new` (six steps incl. “Find your building”), `/p/:place[/b/:building]` (Make, Look, Choose, Prove, Tell at City to Room; desktop ladder and panel, tablet side sheet, phone bottom sheet), `/settings` (ledger with impact before Apply). Data: GB1 to GB3 fixtures and the Hyderabad pack; stages: `@ibasho/preview` compute functions via `src/studio/preview/runner.ts` (rough, refined, final, cancel, generations, only touched stages rerun); scene: `@ibasho/scene` `CityStage` with `focus`; find: header box and Ctrl or Cmd K palette over `@ibasho/plan` `searchPlaces`. **Stubbed.** Non-geometry levers via `levers.ts` `adjustEnergy` (see the WP-08 interface entry to A); jobs on the main thread (worker open); PDF underlay needs `pdfjs-dist`; trace assist, wall chain and doors not built; Room level shows building-wide numbers. **QA round 1 folded in:** place hits, `storeyName`, `places` in the index, `replacedIds` per job, Himalaya block hits focus by `bbox`, `ECSBC.EPI-RATIO`, new ledger rows (demand charge, power factor, Gruha Jyothi, solar hot water). The scene sits in an error guard and routes have an Andon error page. Screenshots: `docs/qa/ux/V-P/studio/` (7). **Tests (each once, capped).** ui 10 files 88 passed; editor 1 file 5 passed; store 1 file 10 passed; web 4 files 22 passed. Typecheck clean for ui, editor, store and apps/web. Biome clean (warnings only); `tools/lint` clean.

**agent-A:WP-00** handoff · done

**QA round 1 (docs/qa/product/platform-V-P.md) fixed by Agent A.** Blockers: B1 GB1 refitted to the OSM outline (rotation 41.7, courts on the right sides, 3 m C spine corridor, test IoU at least 0.8), B2 93 named rooms with function aliases from RB 3.2 (no personal names), B3 ECSBC by EPI ratio to the reference design. Majors M1 to M11 done (HT-II(A) per kVAh with time of day, demand from a load sum, stale generation guard with a test, places and Himalaya blocks in search, unknown heights never shade and the context radius is honoured, replacement by listed ids, GB2 off Kohli Research Block, apartment end uses and seasonal AC, LT-I cliff and Gruha Jyothi, ENS RETV and openable area, LPD per type, daylight only for occupied rooms, threshold margins inside the geometry uncertainty read unknown). Minors done: m1, m2, m3, m4, m6, m7, m8, m9, m10, m13. Left: m5 (set deviceFactor 0 for instant replay), m11 (sample places keep the pack origin so frames match; search centres them on their buildings), m12 (Open Buildings heights at V-2). Tests: core 146, plan 17, preview 26, city-hyderabad 2, tools/city 3; typecheck, lint:tools and depcruise clean. See the QA round 1 interfaces entry.

**builder:WP-07 (agent C)** handoff · done

**City stage, design QA round.** Applied `docs/qa/design/city-V-P.md`: B1 night lanterns (lamplight windows, warm body, end use pools and halos, several Preview lanterns, night city by value, 24 h clock, Lanterns eases to 20:30), B2 phone, tablet and desktop layouts with a chrome safe area, B3 Andon chrome via WP-06 components, B4 framing, 1.5 px ink outline, site plinth and City locator, B5 daylight tray with pass line and share bar from one pass mask, M3 focus colour highlight, M10 shadow hours in a shade ramp, plus parts of M2, M4, M5, M6, M7, M9, M11, m1 to m7. Real data swapped in (pack tiles, compiled GB1, replaced ids, PlaceHit focus). Lens kernels in a worker. Not done: M1 AO and edge bake, M8 sun bent film strip and seal, M12 rose and moment 7, pinch and pan gestures. Scene tests 91 passed.

**agent-A:WP-13+WP-14** handoff · done

**QA round 2 minors and design levers (Agent A).** R2-m1 `SiteResult.unknownHeightIds`. R2-m2 AC hours now differ by set point (GB1 first room: 2,454 to 2,727 h at 24 C, 2,212 to 2,458 h at 26 C). R2-m3 search: wing codes (“A3”, “C7”) and Bodh room numbers as aliases, cafeteria rooms, exact names and aliases rank above space-type words and block parts (“classroom” finds Classroom CR-1 first). R2-m4 codes and baselines follow the place’s code packs: GH1 London has no ECSBC baseline, only the UK EPC estimate row. R2-m5 GB2 cooling 899 MWh, about 2,170 full-load hours on the 400 TR plant (ASHRAE 62.1 outdoor air plus 0.3 air changes instead of 1.5, people at 60% diversity, blinds, academic term for classrooms and halls). R2-m6 GB1 end wings split to their labels (EERC about 193 m², SBI about 200 m², the rest as stair and service cores), 99 rooms. Levers: see the WP-13 interface entry. Regenerated building and Preview fixtures. Tests, one at a time with maxWorkers 1: plan 19, preview 31, core 146 passed; typecheck core, plan, preview clean; Biome and lint:tools clean. Typst not run (orchestrator compiles).

**builder:WP-06+WP-08+WP-15 (agent B)** handoff · done

**Studio review round 2 (docs/qa/design/studio-V-P.md), agent B items.** B-B1 active chips are ink on `--paper-sunk` with a 2 px bead (contrast test). B-B2 studio jobs run in a dedicated worker (`apps/web/src/studio/worker/studio.worker.ts`, client `preview/jobs.ts`; tiles sent once; in page fallback without workers); rough outputs paint at once (`Job.latest`). B-B3 the ledger counts from the stage graph (`stageQuantities`, `downstreamStages`), “Working out what this changes” while computing, never 0. B-B4 fluid headings, `overflow-x: clip`, tablet sheet between top bar and engawa, narrow crumbs keep the building, phone New flow shows “Step n of 6”. B-M1 the studio shows one engawa line (“Since your last visit: new tariff, ECSBC tiers, …”), dismiss remembered. B-M2 rough site results paint the scene. B-M3 legend facts as a definition list, legend top covers every value, facade histograms gone. B-M4 phone panel uses `ModeTabs`; compact stage bar shows the stage name; short stage names (Geometry, Sun, Radiation, Daylight, Energy, Codes, Search). B-M5 the plan is drawn true north up like the 3D (labels upright), one label rule (function, then name, 11 px minimum, falls back to the short function), labelled steppers (“Floor to floor”, “Copies of this floor”), facade slider and chip read the same value, ghost only for an above-ground storey below, room for the rail. B-M7 comfort copy by type and IMAC in plain words. B-M8 `levers.ts` deleted: design levers go to the Preview energy stage as `EnergyParams` (`preview/variant.ts`, `PARAM_PRESETS`, `pvKwpFor`). B-M9 tiles no longer overflow their captions, “Main view” and “Show large”, the building as it is runs first. A-M3 every headline (home revisit, settings impact, studio) comes from one function over the job’s own outputs. R2-m0 test: a late older generation never replaces the live job. Minors m1, m2 (Not found page), m3, m5, m6, m7 (memory and lever copy as sentences), m8, m13 (time strip only in Look). QA round 1 folded: places in the search index, `replacedIds`, `ECSBC.EPI-RATIO`, new money and comfort settings in the ledger, block hits focus by box, place hits. **Left:** B-m9 unmount warning, B-m10 search grouping cap, B-m11 home card menu, B-m12 font preload, compare difference drawing (needs C). **Tests (each once, capped, one worker):** web 4 files 23 passed, ui 10 files 89 passed, editor 5 passed; typecheck and Biome clean on owned paths.

**builder:WP-08+WP-15 (agent B)** handoff · done

**Design before you build (SPEC 27) and the customer-facing sweep.** Route `/app/design` (`apps/web/src/studio/pages/DesignFlow.tsx`, top bar link “Design”): Brief (type cards, name, area, storeys, budget band, goals as typed `Goals`), Design (`PlanEditor`, `StoreyStack`, `FacadePanel`, start from a type template or as massing, “Add a stair and lift core” via new editor recipes `addCore`, `removeCore`, cores drawn hatched), Place (Hyderabad or London; click open ground on `CityStage` (`onSelect` ground), nudge buttons, a `Range` to turn it, overlap check against mapped footprints), Compare (up to three plots or orientations, each run in the studio worker with its own stage bar, EUI, bill, carbon, roof sun, daylit floor and savings), Shape to goals (`goalGaps`: met, not met,

unknown or not applicable with its source, never compliant; optimiser picks applied as design levers), Commit (a new place and building with a “Designed before building” version and a “Shaped to the goals” what-if). Waiting on: agent C’s placement controls in `CityStage` (dragging; clicking ground works now), agent A’s goal targets and UK code pack (Part L reads unknown until then). **Customer-facing sweep:** sources and notes in plain words (Telangana commercial tariff, Typical weather year for Hyderabad, Bureau of Energy Efficiency, Google Open Buildings heights), no pack, tier, fixture or stage wording on screen, the dev “drop a pin” block removed. **Tests:** web 4 files 24 passed (adds the goal gap test), ui 89, editor 5, store 10; typecheck and Biome clean; `tools/lint` clean.

**builder:WP-07 (agent C)** handoff · done

**Studio review (S-B1, S-M1, S-M2, S-M3, B-M5 share) and SPEC 27 placement, scene side.** Hero is the brightest lantern at every level (warm body, lamplight windows, pool and halo), neighbours capped, context dark with dashed outlines, sun lenses yield to Lanterns at night. Penumbra limited to neighbours that reach the hero, dashed tops elsewhere, legend line “Dashed tops: estimated heights”. Daylight drawer slides to the side with the fewest neighbours and the city fades to 30% while it is open. Building and Floor framing fit the whole building. Room labels fit or collapse, function first. Placing: drag, turn handle, steps, ground level, overlap and shade check drawn in the scene (warm and focus outlines) and listed in a Place it panel; `CityCompare` for plots and bearings side by side. Customer wording sweep: no stand in, placeholder or stage jargon in the stage; fallbacks read “Approximate neighbourhood” and “Example building”. Kernels for site and sun hours run in a module worker. Tests: scene 13 files, 96 passed (vitest, one worker). Not done: M1 AO and edge bake, M8 film strip and seal, M12 radiation rose and moment 7, pinch and pan.

**builder:WP-08+WP-15 (agent B)** handoff · done

**Folded agent A and agent C landings into the studio.** Bills on every screen come from `headlineBill` (views), so Vindhya’s bill matches across Home, Settings and the studio. Design before you build reads `GOAL_PRESETS` (labels in plain words, UK or India marked), passes the brief’s goals to the codes stage (`StageIn.codes.goals`) and shows `goalGap` results per goal with value, target and source (`designGaps`); London designs use `pl_kings_cross`. Placing uses `CityStage` `placeMode`, `placement`, `onPlacementChange`, `onSiteCheck` (overlaps and who shades it, in words), and the compare step shows `CityCompare` panes above the numbers; `hideContextIds` is always passed. Make’s 3D beside the plan uses `azimuthDeg={180}` (both north up). Worker-side modules import the new `@ibasho/editor/core` entry (geometry, checks, commands, room functions, presets; no React). **Open:** `heatPumpScop` lever, on-demand `CITY_AREAS` loading for searched places, B-m9 to B-m12. Round 3 check shots: `docs/qa/ux/V-P/studio/round3/` (look 1280, choose 1600, make 1600, phone 360, settings 360, design 1280; no page errors, no sideways scroll); in look 1280 the facades are not yet painted under Radiation although `site` is passed (worth a look by agent C).